JEE Mains 2019 Chapter wise Question Bank

Calorimetry - Questions

Q1

An unknown metal of mass 192 g heated to a temperature of 100°C was immersed into a brass calorimeter of mass 128 g containing 240 g of water at a temperature of 8.4°C. Calculate the specific heat of the unknown metal if water temperature stablizes at 21.5°C. (Specific heat of brass is 394 J kg^{-1} K^{-1})

(1) 458 J kg^{-1} K^{-1} (2) 1232 J kg^{-1} K^{-1}
(3) 916 J kg^{-1} K^{-1} (4) 654 J kg^{-1} K^{-1}

10 Jan Evening

Q2

Ice at –20°C is added to 50 g of water at 40°C, When the temperature of the mixture reaches 0°C, it is found that 20 g of ice is still unmelted. The amount of ice added to the water was close to
(Specific heat of water = 4.2J/g/°C
Specific heat of Ice = 2.1 J/g/°C
Heat of fusion of water at 0°C = 334J/g)

(1) 50g (2) 100 g
(3) 60 g (4) 40 g

11 Jan Morning

Q3

When 100 g of a liquid A at 100°C is added to 50 g of a liquid B at temperature 75°C, the temperature of the mixture becomes 90°C. The temperature of the mixture, if 100 g of liquid A at 100°C is added to 50 g of liquid B at 50°C, will be :

(1) 85°C (2) 60°C
(3) 80°C (4) 70°C

11 Jan Evening

Q4

A metal ball of mass 0.1 kg is heated upto 500°C and dropped into a vessel of heat capacity 800 JK^{-1} and containing 0.5 kg water. The initial temperature of water and vessel is 30°C. What is the approximate percentage increment in the temperature of the water? [Specific Heat Capacities of water and metal are, respectively, 4200 $Jkg^{-1}K^{-1}$ and 400 Jkg^{-1} K^{-1}]

(1) 15% (2) 30%
(3) 25% (4) 20%

11 Jan Evening

Q5

A thermally insulted vessel contains 150 g of water at 0°C. Then the air from the vessel is pumped out adiabatically. A fraction of water turns into ice and the rest evaporates at 0°C itself. The mass of evaporated water will be closed to :
(Latent heat of vaporization of water = 2.10×10^6 J kg^{-1} and Latent heat of Fusion of water = 3.36×10^5 J kg^{-1})

(1) 150 g (2) 20 g (3) 130 g (4) 35 g

8 April Morning

Q6

A massless spring (K = 800 N/m), attached with a mass (500 g) is completely immersed in 1kg of water. The spring is stretched by 2cm and released so that it starts vibrating. What would be the order of magnitude of the change in the temperature of water when the vibrations stop completely? (Assume that the water container and spring receive negligible heat and specific heat of mass = 400 J/kg K, specific heat of water = 4184 J/kg K)

(1) 10^{-4} K (2) 10^{-5} K (3) 10^{-1} K (4) 10^{-3} K

9 April Evening

Q7

When M_1 gram of ice at $-10\ ^oC$ (Specific heat = 0.5 cal $g^{-1}\ ^oC^{-1}$) is added to M_2 gram of water at 50^oC, finally no ice is left and the water is at 0^oC. The value of latent heat of ice, in cal g^{-1} is:

(1) $\frac{50M_2}{M_1}-5$ (2) $\frac{5M_1}{M_2}-50$

(3) $\frac{50M_2}{M_1}$ (4) $\frac{5M_2}{M_1}-5$

12 April Morning

Q8

One kg of water, at 20^oC, is heated in an electric kettle whose heating element has a mean (temperature averaged) resistance of 20 Ω. The rms voltage in the mains is 200 V. Ignoring heat loss from the kettle, time taken for water to evaporate fully, is close to :

[Specific heat of water = 4200 J/(kg^{o}C), Latent heat of water = 2260 kJ/kg]

(1) 16 minutes (2) 22 minutes

(3) 3 minutes (4) 3 minutes

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Calorimetry - Answers

Q1

(3) Let specific heat of unknown metal be 's' According to principle of calorimetry, Heat lost

$= \text{Heat gain } m \times s\Delta\theta = m_1 s_{brass} (\Delta\theta_1 + m_2 s_{water} + \Delta\theta_2)$

$\Rightarrow 192 \times S \times (100 - 21.5)$

$= 128 \times 394 \times (21.5 - 8.4)$

Solving we get,$+ 240 \times 4200 \times (21.5 - 8.4)$

$S = 916 \text{ Jkg}^{-1}\text{k}^{-1}$

10 Jan Evening

Q2

(4) Let m gram of ice is added.

From principal of calorimeter

heat gained (by ice) = heat lost (by water)

$\therefore\ 20 \times 2.1 \times m + (m - 20) \times 334$

$= 50 \times 4.2 \times 40$

$376\ m = 8400 + 6680$

$m = 40.1$

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Q3

(3) Heat loss = Heat gain = $mS\Delta\theta$

So, $m_A S_A \Delta\theta_A = m_B S_B \Delta\theta_B$

$\Rightarrow 100 \times S_A \times (100 - 90) = 50 \times S_B \times (90 - 75)$

$2S_A = 1.5S_B \Rightarrow S_A = \frac{3}{4} S_B$

Now, $100 \times S_A \times (100 - \theta) = 50 \times S_B \times (\theta - 50)$

$2 \times \left(\frac{3}{4}\right) \times (100 - \theta) = (\theta - 50)$

$300 - 3\theta = 2\theta - 100$

$400 = 5\theta \Rightarrow \theta = 80°C$

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Q4

(4) Assume final temperature = T°C

Heat lass = Heat gain = $ms\Delta T$

$\Rightarrow m_B\, s_B\, \Delta T_B = m_w\, s_w \Delta T_w$

$0.1 \times 400 \times (500 - T)$

$= 0.5 \times 4200 \times (T - 30) + 800\ (T - 30)$

$\Rightarrow 40\ (500 - T) = (T - 30)\ (2100 + 800)$

$\Rightarrow 20000 - 40T = 2900\ T - 30 \times 2900$

$\Rightarrow 20000 + 30 \times 2900 = T(2940)$

$T = 30.4°C$

$\frac{\Delta T}{T} \times 100 = \frac{6.4}{30} \times 100 = 21\%,$

so the closest answer is 20%.

11 Jan Evening

Q5

(2) Suppose amount of water evaporated be M gram.
Then (150 – M) gram water converted into ice.
so, heat consumed in evoporation = Heat released in fusion

$M \times L_v = (150 - M) \times L_s$

$M \times 2.1 \times 10^6 = (150 - M) \times 3.36 \times 10^5$

$\Rightarrow M \simeq 20$ g

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Q6

(2) $\frac{1}{2}.kx^2 = mC(\Delta T) + m_\omega C_\omega \Delta T$

or $\frac{1}{2} \times 800 \times 0.02^2 = 0.5 \times 400 \times \Delta T + 1 \times 4184 \times \Delta T$

$\therefore \Delta T = 1 \times 10^{-5} K$

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Q7

(1) $M_1 C_{ice} \times (10) + M_1 L = M_2 C_{\omega} (50)$

or $M_1 \times C_{ice} (=0.5) \times 10 + M_1 L = M_2 \times 1 \times 50$

$$\Rightarrow L = \frac{50 M_2}{M_1} - 5$$

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Q8

(2)

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